

RAJA KUMAR

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SUMMARY

Final-year B.Tech CSE student (CGPA 9.11) specializing in AI/ML, Generative AI, and intelligent systems. Experienced in building RAG applications, AI agents & vector databases, using Python, FastAPI, Docker, and CI/CD.

SKILLS

Programming Languages: Java, Python, C, SQL

Frameworks & Libraries: FastAPI, Flask, LangChain, LangGraph, Next.js, React, Express, TensorFlow

Tools & Platforms: AWS, Docker, Git, GitHub Actions, VS Code, PyCharm, Jupyter Notebook, CI/CD

AI/ML & Software Engineering: Machine Learning, Deep Learning, Generative AI, LLMs, RAG, System Design

Core Competencies: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

Yseek: Youtube Video RAG Engine | Python, Qdrant, faster-whisper, Sentence-Transformers

 [Github](#)

- Processed **200+** hours of video via **faster-whisper** and **yt-dlp**, engineering a 75s time-window chunker to preserve temporal contexts.
- Built a **Qdrant** retrieval engine indexing **30,000+** vectors, delivering **250ms** latency and **91% Hit@3** accuracy on multilingual queries.
- Integrated **FastAPI** and the **YouTube IFrame API** for instant seek-to-second playback and grounded LLM synthesis.

AI-Map: Local RAG & Vector Database Platform | Python, HNSW, Ollama, Docker, GitHub Actions

 [Github](#)

- Engineered a custom Vector Database in Python using HNSW ($O(\log N)$) and KD-Trees, processing **10,000+** embeddings with **sub-50ms** query latency.
- Deployed an air-gapped RAG pipeline, improving context retrieval accuracy by **40%** for AI-powered question answering.
- Developed a PCA-based visualization dashboard to benchmark **3** distinct search algorithms, reducing performance profiling time by **2 hours** per cycle.
- Streamlined application deployment with Docker Compose, accelerating CI/CD pipeline builds by **25%** via GitHub Actions.

Quant Agent: Multi-Agent Trading System | Python, LangGraph, LangChain, Flask

 [Github](#)

- Architected a multi-agent AI trading platform utilizing LangGraph, automating **100%** of routine market analysis and trade orchestrations.
- Integrated GPT-4o, LLaVA, and TA-Lib indicators (RSI, MACD), reducing false trading signals by 30%.
- Constructed high-throughput Flask REST APIs to handle **50+** concurrent agent requests with **<200ms** inference latency.
- Automated cloud infrastructure deployment via Docker and GitHub Actions, cutting release cycles from days to **under 15 minutes**.

EDUCATION

KIIT UNIVERSITY

2023 – 2027

B.Tech in Computer Science and Engineering

CGPA: 9.11

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Discrete Mathematics, Machine Learning, Data Mining

WORK EXPERIENCE

InAmigos Foundation | *Machine Learning Intern*

Aug 2026 – Present

- Engineered ML training pipelines encompassing systematized data preprocessing, outlier detection, and robust feature engineering across **100,000+** tabular records.
- Benchmarked classification and regression models (XGBoost, Random Forest, LightGBM), elevating model F1-score by **14%** and reducing inference error by **18%**.

Oasis Infobyte | *Data Analytics Intern*

Sep 2026 – Present

- Conducted EDA and RFM feature extraction on **45,000+** consumer records, isolating 4 distinct behavioral customer segments using K-Means clustering.
- Constructed an NLP sentiment classifier using TF-IDF feature matrices and ensemble models, achieving an **89% test accuracy score** on customer feedback.

ACHIEVEMENTS & CERTIFICATIONS

- Earned **3** professional credentials including AWS Cloud Architecting, AICTE Data Science Internship (Siemens), and Data Science, AI/ML & GenAI, covering **100+** hours of advanced curriculum.
- Solved 300+ DSA problems and secured 1st Runner-Up at KIIT Hackathon UiPath Resume Screener.